



Cloud-Based IoT Environmental Telemetry Platform

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Functional Requirements Overview

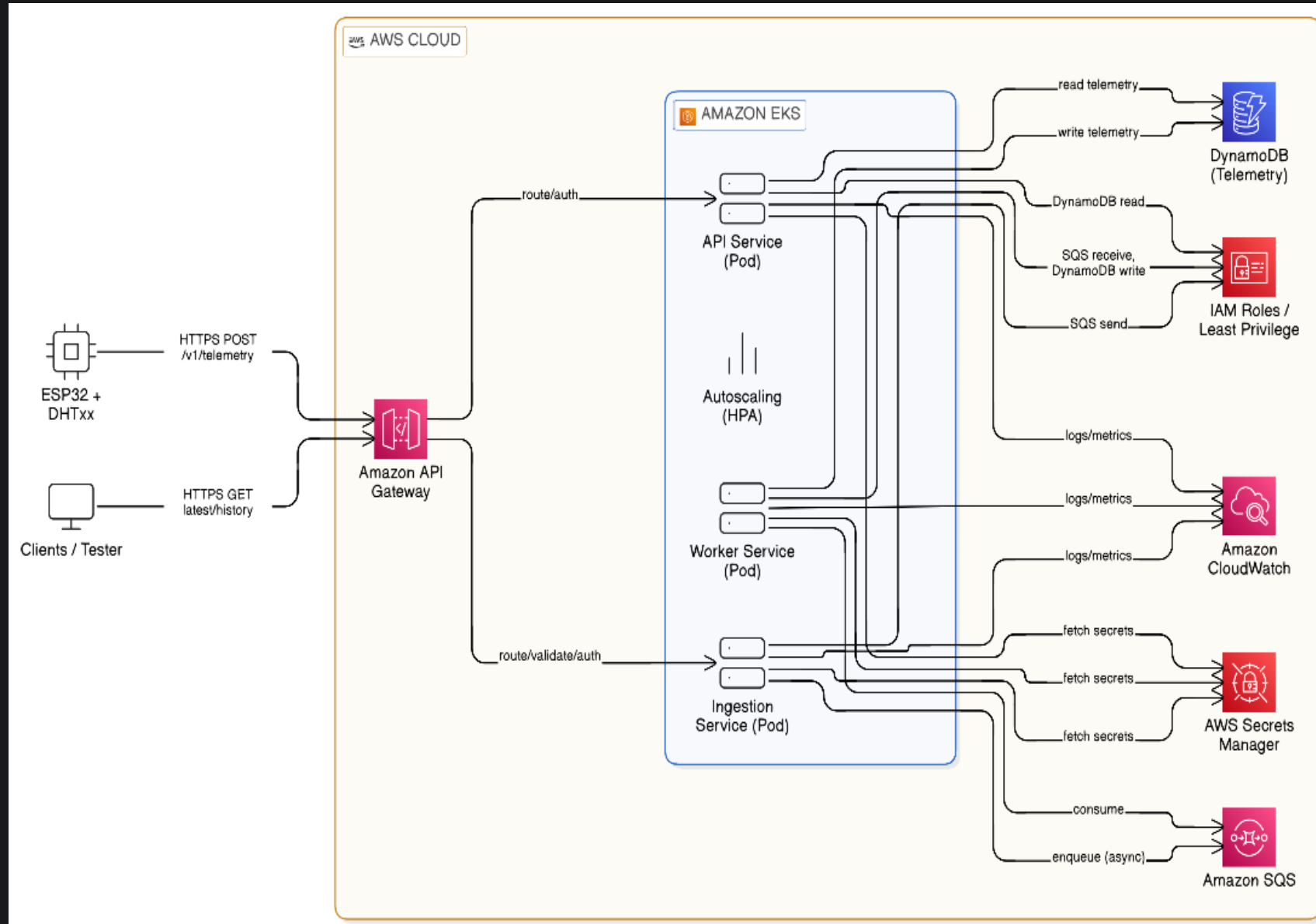
- **Functional Requirements (FRs)**
 - **FR-1** The system shall ingest temperature and humidity data via an API or message stream
 - **FR-2** The system shall persist telemetry data in a cloud database
 - **FR-3** The system shall expose APIs to retrieve latest and historical telemetry
 - **FR-4** The system shall deploy ingestion and API services as independent containers
 - **FR-5** The system shall authenticate API requests
 - **FR-6** The system shall process ingestion asynchronously using a queue or stream
 - **FR-7** The system shall expose health endpoints for services
 - **FR-8** The system shall deploy to Kubernetes with ingress and service discovery
 - **FR-9** The system shall auto-scale services based on load
 - **FR-10** The system shall support rollback to previous deployments



Nonfunctional Requirements Overview

- Non-Functional Requirements (NFRs)
 - NFR-1 Infrastructure must be reproducible via Infrastructure as Code
 - NFR-2 The system must tolerate individual service failures
 - NFR-3 Secrets must be securely managed and never stored in code
 - NFR-4 Logs and metrics must be centrally collected
 - NFR-5 Alerts must be generated for failures or threshold breaches
 - NFR-6 The system must operate within defined cost controls
 - NFR-7 The system must support operational readiness (backup, rollback, runbook)

Architecture Recap



Backlog And Requirements Coverage

Status	Backlog Item	Milestone	Requirements Covered	Implementation / Evidence
Done	Repo & architecture defined (B-1, B-2)	M2–M3	FR-4, NFR-1	Git repo, README, architecture diagram
Done	Ingestion API prototype (B-3)	M4	FR-1, FR-7	POST /v1/telemetry, validation, /health
Done	Query API prototype (B-4)	M4	FR-3, FR-7	GET /v1/devices/{id}/telemetry/latest, /health
Done (Partial)	Telemetry persistence	M4	FR-2	File-backed latest-per-device storage
Next	Observability & security baseline (B-5)	M5	FR-7, NFR-3, NFR-4	Logging, metrics, auth approach, secrets plan

Backlog And Requirements Coverage Continued

Status	Backlog Item	Milestone	Requirements Covered	Implementation / Evidence
Next	Requirements traceability & CDR	M5	All FRs / NFRs	Traceability table, updated architecture
Planned	Kubernetes deployment (B-6)	M6	FR-8, FR-9	Ingress, service discovery, HPA
Planned	Async ingestion + worker (B-7)	M7	FR-6	Queue/stream, worker service, simulated device
Planned	Load testing & cost controls (B-8)	M8	FR-9, FR-10, NFR-6, NFR-7	Load tests, tuning results, rollback, budgets



Prototype Scope

- **Implemented**

- Separate Ingestion and API microservices
- Telemetry ingestion with field and type validation
- /health endpoints for both services
- Container-ready, independently deployable services

- **Simulated/Unimplemented**

- Asynchronous ingestion (queue/stream mocked)
- Cloud database (JSON file used instead)
- IoT devices (manually simulated producers)
- Authentication & authorization
- Autoscaling behavior

Prototype Evidence

```
PS C:\Users\seidc\cs361-iot-cloud-telemetry-platform> Invoke-RestMethod -Method Get -Uri "http://localhost:8082/v1/devices/esp32-001/telemetry/latest"
```

device_id	humidity_pct	temperature_c	timestamp
esp32-001	41.2	22.4	2026-03-02T18:00:00Z

```
PS C:\Users\seidc\cs361-iot-cloud-telemetry-platform> curl.exe -i -X POST "http://localhost:8081/v1/telemetry" -H "Content-Type: application/json" --data-binary "@telemetry.json"
```

```
HTTP/1.1 202 ACCEPTED
```

```
Server: gunicorn
```

```
Date: Thu, 22 Jan 2026 21:13:50 GMT
```

```
Connection: close
```

```
Content-Type: application/json
```

```
Content-Length: 23
```

```
{"message": "Accepted"}
```

```
PS C:\Users\seidc\cs361-iot-cloud-telemetry-platform> curl.exe -i "http://localhost:8082/v1/devices/esp32-001/telemetry/latest"
```

```
HTTP/1.1 200 OK
```

```
Server: gunicorn
```

```
Date: Thu, 22 Jan 2026 21:15:21 GMT
```

```
Connection: close
```

```
Content-Type: application/json
```

```
Content-Length: 102
```

```
{"device_id": "esp32-001", "humidity_pct": 41.2, "temperature_c": 22.4, "timestamp": "2026-03-02T18:00:00Z"}
```


Implementation Plan



M5 – Design Readiness & Baseline

Requirements traceability (FR/NFR → components/tests)

Security baseline (auth approach, secrets management)

Observability baseline (logs, metrics, health endpoints)

Critical Design Review (CDR)



M6 – Kubernetes Deployment

Deploy services to Kubernetes

Configure ingress & service discovery

Implement autoscaling (HPA)

Validate live deployment



M7 – Asynchronous Processing

Add queue/stream and worker service

Simulated IoT publisher/device

Ingestion visibility (API endpoint or dashboard)



M8 – Performance & Operations

Load testing and bottleneck tuning

Cost controls (tags, budgets, cleanup)

Operational readiness check & final demo